Computing Solutions to Water and Sanitation for Sustainable Development (SDG 6)

Design and develop an innovative computing solution that improves access to clean water, sanitation, and hygiene by leveraging digital technologies, data analytics, IoT, mobile platforms, or AI.

The solution should address real-world challenges related to water availability, water quality monitoring, sanitation infrastructure, hygiene awareness, and sustainable water management in urban or rural communities. Participants should focus on scalable and technology-driven approaches that contribute to **Sustainable Development Goal 6: Clean Water and Sanitation for All**.

Objective

Create a functional prototype of a tool that helps communities or institutions to:

1. Monitor and manage water quality and availability in real time
2. Improve sanitation infrastructure planning and maintenance
3. Promote hygiene awareness and behavioral change
4. Optimize water distribution, conservation, and reuse
5. Support decision-making for sustainable water resource management.

Solution Scope

Participants may build tools focused on one or more of the following areas:

1. **Smart water monitoring systems** using IoT sensors for water level, quality, and leakage detection
2. **Water conservation and management platforms** for households, agriculture, and industries
3. **Sanitation infrastructure monitoring systems** for toilets, sewage systems, and waste-water treatment
4. **Hygiene awareness platforms** using mobile apps, gamification, or digital education tools
5. **Data-driven water resource planning** using AI, ML, and predictive analytics
6. **Community reporting platforms** for water issues such as contamination, shortages, or sanitation failures
7. **Smart irrigation and water-efficient agriculture systems**

Target Users

The proposed solution may be designed for:

1. Rural and urban households
2. Municipal corporations and local governance bodies
3. Water resource management authorities
4. NGOs and organizations working in water and sanitation
5. Farmers and agricultural communities
6. Schools and educational institutions promoting hygiene awareness

Evaluation Criteria

Submissions will be evaluated based on:

1. Alignment with **SDG 6 goals and social impact**
2. Innovation and practical applicability
3. Ease of use and accessibility for communities
4. Technical implementation and reliability
5. Scalability and sustainability of the solution

Expected Deliverables

Each team should submit:

1. **A working prototype or demonstration**
2. **A brief description of the water or sanitation challenge addressed**
3. **A technical explanation of the computing solution**
4. **A deployment strategy and potential societal impact**

Bonus Considerations

Additional points may be awarded for:

1. **Low-cost and scalable solutions for rural or resource-constrained regions**
2. **Integration with IoT sensors or real-time data systems**
3. **Mobile-first or offline-capable platforms**
4. **Integration with government initiatives related to water and sanitation**
5. **Use of AI, ML, GIS, or data analytics for water management**